

Proof of conjecture 00000002490:
subspace counts over finite fields are Gaussian binomials,
and so are the counts in every interval of the subspace lattice

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1 The conjecture and its reading

Statement (English, verbatim). Definition: The finite-field side of partitions: q-binomial lattices. Conjecture: The counting of subspaces over finite fields is governed by Gaussian binomial coefficients: the lattice count closes under the Gaussian binomial structure. (partition finite-field side q-binomial lattice Gaussian binomials)

Chinese text. It says, in substance, that on the finite-field side of partitions the counts of the lattice are Gaussian binomials.

This is a classical theorem, not a new result. We prove it and check the proof in Lean 4 with Mathlib. The wording is vague, so we prove the following explicit statements. Each one is a theorem in the Lean file.

- (R1) *Counting.* Let $\mathbb{F}$ be any finite field with q elements and let $n, k \geq 0$. The number of k -dimensional subspaces of $\mathbb{F}^n$ is $\begin{bmatrix} n \\ k \end{bmatrix}_q$. More generally, the same holds for every n -dimensional $\mathbb{F}$ -vector space V .
- (R2) *The counts close under the q-Pascal structure.* Write $N_q(n, k)$ for the number in (R1). Then both q-Pascal rules hold for these counts: $N_q(n+1, k+1) = N_q(n, k) + q^{k+1}N_q(n, k+1)$, and, for $k \leq n$, $N_q(n+1, k+1) = N_q(n, k+1) + q^{n-k}N_q(n, k)$.
- (R3) *Interval closure.* Every interval of the subspace lattice is again counted by Gaussian binomials. Let $U \leq T$ be subspaces of a finite $\mathbb{F}$ -vector space V , and let $\dim U \leq k$. Then the number of k -dimensional W with $U \leq W \leq T$ is $\begin{bmatrix} \dim T - \dim U \\ k - \dim U \end{bmatrix}_q$.

Not formalized. The word “partitions” appears only in the heading. We do not formalize the separate fact that the coefficients of $\begin{bmatrix} n \\ k \end{bmatrix}_q$ count partitions in a box. “Lattice” is read as the lattice of subspaces of $\mathbb{F}^n$, ordered by inclusion.

Conventions. $\mathbb{F}$ is an arbitrary finite field with $q = |\mathbb{F}|$ elements, so $q \geq 2$. Its prime-power form is never used. A subspace is a Mathlib `Submodule`. Dimension is `Module.finrank`. A count is `Nat.card` of a subtype. All $k \geq 0$ are allowed, and for $k > n$ both sides are 0. Natural-number subtraction is truncated, so $n - k$ and $k - \dim U$ appear only under the hypotheses $k \leq n$ and $\dim U \leq k$. We write $\binom{k}{2} = k(k-1)/2$.

2 Definitions as formalized

Definition 1 (`gaussBinom`). The Gaussian binomial $\begin{bmatrix} n \\ k \end{bmatrix}_q \in \mathbb{N}[q]$ is defined by the rules $\begin{bmatrix} n \\ 0 \end{bmatrix}_q = 1$, $\begin{bmatrix} 0 \\ k+1 \end{bmatrix}_q = 0$, and $\begin{bmatrix} n+1 \\ k+1 \end{bmatrix}_q = \begin{bmatrix} n \\ k \end{bmatrix}_q + q^{k+1} \begin{bmatrix} n \\ k+1 \end{bmatrix}_q$.

Wikipedia [W] gives this rule as the first “analog of Pascal’s identity”. It writes the rule as $\binom{m}{r}_q = q^r \binom{m-1}{r}_q + \binom{m-1}{r-1}_q$. Wikipedia *defines* the coefficient by the product formula

$$\binom{m}{r}_q = \frac{(1 - q^m)(1 - q^{m-1}) \cdots (1 - q^{m-r+1})}{(1 - q)(1 - q^2) \cdots (1 - q^r)}.$$

We prove that our definition agrees with that one. Lemma 3 shows that $\begin{bmatrix} n \\ k \end{bmatrix}_q \cdot \prod_{i < k} (1 - q^{i+1}) = \prod_{i < k} (1 - q^{n-i})$ holds in every commutative ring. In $\mathbb{Z}[q]$ the denominator is nonzero, because its value at $q = 0$ is 1 (`qDen_X_ne_zero`). So the product formula determines $\begin{bmatrix} n \\ k \end{bmatrix}_q$ uniquely.

3 Proof

Let $P_x(n, k) = \prod_{i < k} (x^n - x^i)$ (Lean `liProd`).

Lemma 2 (`gaussBinom_mul_liProd`). *In every commutative ring, $\begin{bmatrix} n \\ k \end{bmatrix}_q(x) P_x(k, k) = P_x(n, k)$.*

Proof. The proof is by induction on n , for all k at once. The base cases are immediate, since $P_x(0, k+1)$ has the factor $x^0 - x^0 = 0$. The induction uses $P_x(n+1, k+1) = (x^{n+1} - 1)x^k P_x(n, k)$ and $P_x(n, k+1) = P_x(n, k)(x^n - x^k)$. With these, the q-Pascal rule reduces the step to the identity $(x^{k+1} - 1)x^k + x^{k+1}(x^n - x^k) = x^k(x^{n+1} - 1)$. $\square$

Lemma 3 (`gaussBinom_mul_qDen`). *In every commutative ring, $\begin{bmatrix} n \\ k \end{bmatrix}_q(x) \prod_{i < k} (1 - x^{i+1}) = \prod_{i < k} (1 - x^{n-i})$. Both sides are 0 for $k > n$.*

Proof. The proof is again an induction with the same structure. If $k \leq n$, the step is $1 - x^{k+1} + x^{k+1}(1 - x^{n-k}) = 1 - x^{n+1}$. If $k > n$, the numerator contains the factor $1 - x^0 = 0$. $\square$

Theorem 4 ((R1), `card_subspaces_eq_gaussBinom`). *Let V be a finite $\mathbb{F}$ -vector space of dimension n . Then $\#\{W \leq V : \dim W = k\} = \begin{bmatrix} n \\ k \end{bmatrix}_q(q)$.*

Proof. Each linearly independent k -tuple s in V spans a k -dimensional subspace $\text{span}(s)$. The fibre of this map over a subspace W is in bijection with the linearly independent k -tuples of W (`fibreEquiv`). Such a tuple spans W , because its length equals $\dim W$. Mathlib’s `card_linearIndependent` counts the linearly independent k -tuples in an m -dimensional space as $\prod_{i < k} (q^m - q^i)$ when $k \leq m$. Summing over the fibres gives

$$\#\{s \text{ lin. indep.}\} = N \cdot \prod_{i < k} (q^k - q^i) \quad (\text{card_li_eq}),$$

where N is the number of k -dimensional subspaces. Suppose $k \leq n$. Then the left side is $P_q(n, k)$. By Lemma 2 we get $N \cdot P_q(k, k) = \begin{bmatrix} n \\ k \end{bmatrix}_q(q) \cdot P_q(k, k)$ in $\mathbb{Z}$. Since $q \geq 2$, $P_q(k, k) \neq 0$, and we may cancel. Now suppose $k > n$. Then $N = 0$, and $P_q(n, k) = 0$, so $\begin{bmatrix} n \\ k \end{bmatrix}_q(q) = 0$ as well. $\square$

(R2) follows from (R1). The first rule is the defining recursion, evaluated at q (`subCount_pascal`). For the second rule (`gaussBinom_pascal'`, `subCount_pascal'`), multiply both sides by $\prod_{i \leq k} (1 - q^{i+1}) \neq 0$ and use Lemma 3. This is the second Pascal analog in [W]. Lemma 3 also gives the product formula for the counts (`subCount_product_formula`).

Theorem 5 ((R3), `card_subspaces_interval`). *Let $U \leq T \leq V$ and $\dim U \leq k$. Then $\#\{W : U \leq W \leq T, \dim W = k\} = \begin{bmatrix} \dim T - \dim U \\ k - \dim U \end{bmatrix}_q(q)$.*

Proof. First the case $T = V$ (`card_subspaces_above`). The maps $W' \mapsto \pi^{-1}(W')$ and $W \mapsto \pi(W)$ are inverse bijections. Here $\pi : V \rightarrow V/U$ is the quotient map, W' ranges over subspaces of V/U , and W over subspaces containing U (`aboveEquiv`). By rank-nullity, $\dim \pi^{-1}(W') = \dim W' + \dim U$ (`finrank_comap_mkQ`). Apply (R1) to V/U , which has dimension $\dim V - \dim U$. For general T , the maps given by pulling back and pushing forward along $T \hookrightarrow V$ identify the interval $[U, T]$ in V with the subspaces of T that contain U (`intervalEquiv`). These maps preserve dimension. $\square$

4 Lean formalization

The file is `lean/Conjecture2490/Basic.lean`, namespace `C2490`, 365 lines. Its only import is `Mathlib`. The main theorem `conjecture_2490` is a conjunction. It states (R1) for $\mathbb{F}^n$ (`Fin n` $\rightarrow$ $\mathbb{F}$) for all n, k , both rules of (R2), and (R3) for every finite $\mathbb{F}$ -vector space V , for an arbitrary finite field $\mathbb{F}$. The supporting theorems are `card_subspaces_eq_gaussBinom`, `subCount_eq`, `subCount_pascal`, `subCount_pascal'`, `subCount_product_formula`, `gaussBinom_mul_qDen`, `qDen_X_ne_zero` and `card_subspaces_interval`. As a sanity check, an example confirms that $N_2(4, 2) = 35$.

Axiom audit. `lake build` succeeds. `#print axioms` reports [`propext`, `Classical.choice`, `Quot.sound`] for `conjecture_2490`, `card_subspaces_interval`, `subCount_product_formula`, `qDen_X_ne_zero` and `gaussBinom_mul_qDen`. The file contains no `sorry` and no `native_decide`, and it adds no axioms.

Related package. Conjecture 00000002478 has a separate package. It covers Gauss's q -binomial expansion $\prod_{i < n} (1 + q^i t) = \sum_k q^{\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q t^k$ and its counting interpretation. This package covers the subspace count itself, both Pascal rules, and interval closure in the subspace lattice. It does not treat the expansion.

Sources

[W] Wikipedia, “Gaussian binomial coefficient”, https://en.wikipedia.org/wiki/Gaussian_binomial_coefficient (retrieved 2026-10-05). It gives the product-formula definition and the two analogs of Pascal's identity. It also states that $\begin{bmatrix} n \\ k \end{bmatrix}_q$ “counts the number of k -dimensional vector subspaces of an n -dimensional vector space over $\mathbb{F}_q$ ”.

Mathlib: `card_linearIndependent`, in `Mathlib/LinearAlgebra/Matrix/GeneralLinearGroup/Card.lean`.